

k nearest neighbors

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May 24, 2022

1 Introduction

Standard learning framework: $X, Y \in \mathbb{R}^d \times \mathcal{Y} \sim \rho$.

Training data $\{X_i, Y_i, i = 1, \dots, n\}$ are IID drawn from ρ .

- Regression: $\mathcal{Y} = \mathbb{R}^p$, regression function $m(x) = \mathbb{E}(Y|X = x)$.
- Classification: $\mathcal{Y} = \{1, \dots, p\}$
Binary case $\mathcal{Y} = \{0, 1\}$, Bayes decision function $m(x) = \mathbb{1}_{\mathbb{E}(Y|X=x) > 1/2}$.
- Functional regression: $\mathcal{Y} = \mathcal{H}$ could be any Hilbert space.

Idea: approximation by the nearest k data points, implicitly assume that $m(x)$ has some continuity (Cover, 1968).

2 Regression

For any $x \in \mathbb{R}^d$, arrange the observations in an ascending order of $d(x, X_i)$ as $\{X_{(i)}, Y_{(i)}, i = 1, \dots, n\}$.

The original estimator is

$$\hat{m}(x) = \frac{1}{k} \sum_{i=1}^k Y_{(i)}. \quad (1)$$

Generalization of re-weighted k NN is

$$\hat{m}(x) = \sum_{i=1}^k v_i Y_{(i)}.$$

E.g., $v_i = \frac{K(x, X_{(i)})}{\sum_{i=1}^k K(x, X_{(i)})}$, where K is a kernel function.

A further generalized version:

$$\hat{m}(x) = \sum_{i=1}^n v(x; X_1, \dots, X_n) Y_i.$$

Remark 1. k could be a fixed sequence of n , or a random variable, or data-dependent.

Remark 2. *Ties. $d(x, X_i) = d(x, X_j)$ is annoying and may destroy the consistency without proper tie breaking methods. Assume X has density can avoid this problem. Also, assume X in a Polish space and Y in a separable Hilbert space could avoid the problem of measurable function.*

We only consider Eq. (1) in the following.

2.1 Consistency

Consider the L_1 convergence of $J_n = \int_X |m(x) - \hat{m}(x)| \rho(dx)$.

Theorem 1 (Devroye (1982)). *If $k \rightarrow \infty$ and $k/n \rightarrow 0$, then for all ρ with $|Y| \leq M < \infty$, $J_n \rightarrow 0$ in probability.*

Remark 3. *Since Y is bounded, L_1 convergence implies L_p convergence. $k \rightarrow \infty$ ensures we select enough values around x , while $k/n \rightarrow 0$ is necessary to control the bias.*

Theorem 2 (Devroye et al. (1994)). *If $k/\log n \rightarrow \infty$ and $k/n \rightarrow 0$, then for all ρ with $\mathbb{E}Y < \infty$, $J_n \rightarrow 0$ in probability.*

2.2 Rate of convergence

No universal rate for all distributions. Have to restrict to some function classes.

Assumption 1. *m is bounded with α Hölder-smoothness. That is, there exists positive constants B and C such that*

$$|m(x)| \leq B, \quad |m(x) - m(x')| \leq C\|x - x'\|^\alpha.$$

Assumption 2. *$Y = m(X) + \epsilon$, with noise satisfying $\mathbb{E}\epsilon^r < \infty$ for some $r > 2$.*

Assumption 3. *$k/\log n \rightarrow \infty$ and $k/n \rightarrow 0$.*

Theorem 3 (Lian (2011)). *Under Assumptions 1-3, then we have*

$$|m(x) - \hat{m}(x)| = O([\phi^{-1}(2k/n)]^\alpha + (k/\log n)^{-1/2})$$

almost surely, where $\phi(h) = \text{pr}(x' : \|x - x'\| \leq h)$ is so called small ball probability.

Remark 4. *The first term is the bias, while the second term comes from the variance of ϵ .*

Remark 5. *Take $X \sim U([0, 1]^p)$ for example, $J_n = O((k/n)^{\alpha/d} + (k/\log n)^{-1/2})$.*

If we use the L_2 loss that $J_n = \int_X |m(x) - \hat{m}(x)|^2 \rho(dx)$, we know the optimal rate for bounded X is $J_n = O(n^{-2\alpha/(2\alpha+p)})$, when $k = O(n^{2\alpha/(2\alpha+p)})$ (Györfi et al., 2002).

3 Classification

Please refer to Devroye et al. (2013).

References

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